[00:00] 发言人 1: Introduce you to a little bit of that as well, just called expecting that. But the whole differences is instead of just playing again, it’s asking how good it is. And in alcohol, they use a new member to do that.

[00:14] 发言人 1: Okay, so right now in the late2020s beginning and2022, people have been starting to. Writing computer science algorithms to repeat the humans on like focus on some of these imperfect games, right? You do not know what’s the information was a card holding on the outer books.

[00:36] 发言人 1: Okay, so you do take a look at the train is is going like this way, right? So people wants to develop AI algorithm. One of the very important focus in AI community is to develop AI algorithms that are competing.

[00:49] 发言人 1: As humans on game playing, okay? So if you abstract this out, game playing is basically competing against the other agents, okay? And.

[01:00] 发言人 1: The whole focus for today, okay? So let’s... um... let’s define this a little bit more formally. So today we will be looking at a game called Adversary.

[01:12] 发言人 1: Okay? So you are in the competing situations. Okay? So if we have to define this game, okay? We can think about many axes on how these game can be defined.

[01:24] 发言人 1: For example, we can have stochastic versus deterministic. Okay, deterministic games is like you are playing, for example, the chess and the goal is more deterministic games, right? There’s also some kind of stochastic where the game involves.

[01:44] 发言人 1: For example，throwing a dice，like monopoly，couple，so you need to throw a dice，depending on the outcome，you need to do something，right? And some game has one or two，and even more players，right? Like a video game，most likely you have more。Player at start graph, okay? And another very important type of game is called zero sum.

[02:07] 发言人 1: So if you study game theory, this will come up quite often. Basically what it means is that we only have one... So one opponent winning the average everything you lose, so their utility sum together equal to zero. Okay, so that’s a pure combination.

[02:24] 发言人 1: Okay, we will elaborate this a little bit more later. Then the other axis is how much information you can see. Okay, in some games you see the whole thing, like the pack of cases you see the whole state, you see every dog, the location of every dog, you see the location of those et cetera.

[02:41] 发言人 1: But there’s some games where you do not see. Exactly, Docker, right? Let me play holder, so you do not see what kind of cars you have in your hands.

[02:53] 发言人 1: Okay, so these will have to be changed, the type of game, change complexity for you to develop. Effective algorithm, okay? So like I said, in the end what we want is just like a search algorithm.

[03:10] 发言人 1: In the end we want some solution, we want some algorithm which can basically tell you what to do at any given state. Okay，and that’s we call a

solution or a algorithm for this case。Okay，so now if you look at one of the type of games we have been looking at so far，is or deterministic games，right?

[03:32] 发言人 1: There’s many，many different kinds of formulations，but the following that the formulation I’m giving you right now follows more naturally。From the search algorithm we have been seeing so far, okay? So if we try to use the search algorithm search kind of formulation to formalize a deterministic games, then you can be formalized as follows, okay?

[03:56] 发言人 1: Just like search, what you deterministic game, you also can have. States, okay? Imagine the space to be just a particular configuration of the world.

[04:06] 发言人 1: Okay, so if you’re thinking about this as Pac-Man, then it will be a particular location of your Pac-Man, the location of the dogs and seconds. Okay, then you may have some players. Okay, and generally speaking, you can have one to many many players.

[04:22] 发言人 1: And the one important thing is that the game we consider so far. Other games where you are older. Take terms, so when they apply action, they actually apply the action in terms.

[04:36] 发言人 1: Okay, not like many video games where simultaneous we apply actions. In here we are thinking about games where the players make actions in terms. Okay, so I make a move, then you make a move, then I make a move again.

[04:52] 发言人 1: And we take turns. Then just like the successive function of the searching problem, we can have some. Actions, okay? So every player can take some action, and what action is available depends on the state.

[05:07] 发言人 1: That’s what then like in the successive function upper you take an action, the war is going to change, and in a deterministic game this is captured by a transition function. Okay，so later on in the mark of decision processes，when you learn reinforcement learning，you will see this helping up more often。So basically, if you are sitting，imagine now you are sitting at a particular state S。

[05:34] 发言人 1: Okay，and one of the player is applying an action to this particular space，and this will result a change in the space，and this transition function is basically tells you if you are sitting at state s，if you apply action a，how will this state change，what’s the resolving space? Okay，this is captured。By the transition function, okay?

[06:02] 发言人 1: So later on you will see that sometimes in the real world problem solving, how do you model this transition function becomes very, very important. Okay? So now the terminal test basically is the goal test of your search problem, and you can have just a boolean variable.

[06:19] 发言人 1: I mean the boolean outcome. So if you provide a state. This terminal test is going to test if the state is what you want or not. So if you return true, to say that you are target.

[06:32] 发言人 1: So the name is finished, you have what you want or the resource is what you want, or you return false means that you want to give. Okay, so I made decision to work on. Then finally you will have some.

[06:48] 发言人 1: Utility，okay，so basically the terminal ability means that the final utility you get in the end，okay，and this depends on the sequence of action associated to one particular player。Okay，and if you apply this to every state we have seen so far，in the end you will have some report。Okay，so basically the terminal utility is the final utilities for a player who take actions at every given space in the end，was to resolving reward or consequences。

[07:24] 发言人 1: Okay，so this is one of the ways how we can formally or mathematically define our deterministic case。Okay，so given the definitions of a game，we can now define a solution to this game。Okay，then the solution for a given player of the game is a policy。

[07:47] 发言人 1: Okay，so hold on to this in policy for right now。You can imagine this to be a，you can just simply imagine this to be a strategy，which we will divide it more completely。But in the terminology of RL, they usually call this the policy.

[08:05] 发言人 1: Okay, and this is usually a high and using the learnable parameter data. Okay, so basically what it means is now the solution to your game is a function. Just simply remember that to be a function of strategy which you input is there.

[08:24] 发言人 1: Your input is safe, it tells you this strategy, this policy tells you what action you should take. Okay, you tell it which figure in it just tells you this policy or this strategy tells you what action you should take. Okay, in the end supposed to maximize your utility.

[08:47] 发言人 1: Okay，so this is a solution to a game. Okay，so if you imagine this to be original definition of a search problem，in that case the solution is a sequence of actions。Here it is sort of similar to that, right?

[09:02] 发言人 1: So this solution is some kind of strategy, some kind of function. As soon as you provide what state you’re in, it will immediately tell you what action should be taken. Okay，so this is the definition of the deterministic game，and this is what we need by a solution to this game。

[09:20] 发言人 1: Okay，so from the series side，we have the zero-sum game，one of the very. Very important game where you can imagine in this type of game, the agents are having the two players are having completely opposite utilities. Okay, so you can imagine one agent is maximizing some value and the other player is minimizing that value.

[09:50] 发言人 1: So in the end, the total utility for all the players is always equal to zero. Okay，so as you can imagine，this situation。We are in the completely competing situation, so wherever you gain your opponent has to sacrifice, wherever you lose your opponent will immediately gain.

[10:12] 发言人 1: Okay, so this is many of these pure competition games, and this is one of the. topics for today which is to be called as animal stereo. ok 对抗 everywhere we are competing competing ok so pure competitions and in more general cases.

[10:33] 发言人 1: Your game does not have to be all the times competing. You can have some players who are basically your teammates who try to cooperate with you and try to jointly maximize your total reward. Okay, and you will have some games where

at some part of the game you are competing and at some part of the game you are cooperating.

[10:57] 发言人 1: Okay，so all these are possible，but in this. Lecture for this course is concerned, we are interested in adversary the zero sum game where we are playing against the opponent who try directly competing with you at all times. Okay，and we call this adversarial game。

[11:17] 发言人 1: Okay，means we have opponent who try to compete with you all the times，wherever you win，your component go to this，he’s not going to cooperate this。Okay，the total reward for the for you and your player together will be. Okay, so this is the setup for this adversarial search problem.

[11:41] 发言人 1: Any quick questions before we look at how we can have this solved? Okay, so the next part of today’s lecture concerns about how you actually solve. Okay, so let’s just imagine when we like in the search problem in the.

[12:00] 发言人 1: SEARCH PROBLEM AFTER WE DESCRIBE THE PROBLEM, THE NEXT THING WE SHOW YOU IS THE SEARCH TREE, RIGHT? SO HOW YOUR SEARCH ACTUALLY STEP BY STEP PROGRESS? SO WE CAN USE THIS SIMPLE ACTIVITY GAME AGAIN TO SHOW YOU HOW THIS ACTUAL GAME KIND OF AS YOU PLAY HOW THE SEARCH TREE IS GOING TO GROW.

[12:20] 发言人 1: OKAY! SO THIS IS A VERY SIMPLE GAME. OKAY, YOU ONLY HAVE. ONE TWO THREE FOUR FIVE, five different locations where your backpack can be, and one of the locations have a ball.

[12:31] 发言人 1: That’s it okay? So at this particular space you can only take two actions right? In fact you can only go left or right. That’s. Okay，so if you go left, then this game will have a transition function which tells you，okay，if you go left, this is the next day，how the next day will look like，if you go right，if you go right。This is what the next day looked like.

[13:00] 发言人 1: Right again, from this two states you can again go left or go right, and you will know how this state is going to change after you fly the action that we run. Let me say it’s very similar to the previous search okay? But on top of this we want to define a value which is very very important.

[13:24] 发言人 1: Which we call the value of a state, we usually use letter V to denote that. V of us so for those who have been working with or have exposed to reinforcement learning will see this very often. What’s called value function? It actually means the value of the state.

[13:45] 发言人 1: So let me explain to you what it is. Okay, this is one of the most important concept for all the real most learning for all the games. Okay, so the value of the state is device and the balls.

[13:58] 发言人 1: Okay, so let’s say. I want to refine the value, maybe use some different example with these. So let’s say I want to find out the value for this particular state.

[14:13] 发言人 1: Okay，so what it means is that if the game. You begin scars at this state. What is the absolute best outcome that is achievable? If you will use your games project, okay?

[14:33] 发言人 1: So for example, you can play the game starting from here. You can apply all kinds of actions at any point. In the end, how in the end means what’s the game asset, what’s the game and.

[14:48] 发言人 1: You look at the toll, you look at the terminal utility. How will all these possible terminal utilities? What is the best you need? And that best achievable utility. This would define as the value of this.

[15:08] 发言人 1: Okay，so a higher level intuition of this concept is basically the value of the state，simply speaking，is how good or how bad this state is，okay? And that is qualified by saying if your game starts。From this video space.

[15:27] 发言人 1: What is the best possible outcome? What’s the best possible reward you can get in the end? That is so is defined by its value. Okay，so as the first thing may come to your mind is this is going to be really hard to divide，this is going to be really hard to find，because you need to enumerate every possible scene，every possible outcomes。

[15:59] 发言人 1: Once your game starts from here, okay? Which is true, okay? So you actually from this day you need to factor out all the possible children and you need to go to the end.

[16:13] 发言人 1: Get internal rewards and find out their value. If you’re at a terminal state, for example, your game answered here, then you can do that. Okay, so this example basically shows us, so for every movement.

[16:29] 发言人 1: you have a minus one report. For every action, you have minus one more. So you go left, you have minus one.

[16:38] 发言人 1: Go right, you have minus one. So if you eat the box. You have plus ten reward. Okay, so and once you eat the ball, the game starts. Okay, so.

[17:01] 发言人 1: So if you take a look at the scheme tree, if you apply an action of going right. So your reward coming from zero to negative one. 然后 apply going right action again。You are likely to. Then you, but at the same time you also eat the ball of a plus fan, so total reward you in the end you get this plus eight.

[17:30] 发言人 1: Okay，so if you start from here，then you can enumerate again, going left，going right，going left，going right，in the end what I get，including active as well，okay，and we take this max of all these to be the value of this process。Okay，so like I said，for any terminal state，this V of S is known，and for any. In non-terminal states, you actually need to iterate through all its children nodes, all the possible children nodes, and find out that.

[18:11] 发言人 1: Okay，so you actually need a popular o tree in order to find。Any questions on this? Later on，you will see RL and all these are basically how you manipulate this value function。

[18:28] 发言人 1: Okay，so now this is a single agent tree，and how we find it out right now。Remember，for this class we are talking about every series，so there’s two player，the other one is trying to compete with you，so the game changed a little bit。So let’s say you are Pac-Man。Then your opponent is this ghost who try to compete with you.

[18:52] 发言人 1: And remember in our studies, we make terms the player make terms in ultimate so the pattern move first then the ghost move okay need to last time previously so in here. Let’s say your current member first, you can move either to the left or to the right, and these are resulting states. Okay, and we are using the color to be know who’s turn.

[19:22] 发言人 1: Okay, when you see the box with blue, our liner, that means this is the pack that’s turn to make move. The red that makes this be be those profitable. So now from here, your goals can either move left or move right or move move left or move right.

[19:38] 发言人 1: OK, these are all the possibilities. And in the end you might have some report. Okay, for every movement, like if one be the loss was ten getting eaten by the ghost to minus twenty, for example, and in the end you have some values.

[19:52] 发言人 1: Okay, then something change, what changes is now how you define the value for every state because that state. That belongs to different players. Okay, so now instead of simply calling a value of state, now we are looking at what we call the meet mask.

[20:10] 发言人 1: Okay, so what it means is if the states. It’s actually belongs to the main agent or belong to yourself, the one trying to maximize the rewards. Okay, like the pattern. Okay, then the value is defined at what’s the best possible outcome that’s achievable, the same as the previous value of this. Okay，so that is defined as the best power。

[20:41] 发言人 1: Okay，now if you looking at the value for the state that associated with your opponent，like in this case associated with proposed，then the value for AR states is actually the worst that’s achievable of the states。Because remember, we assume your opponents trying to compete with you completely. So the best outcome with your opponent is the worst that is achievable playing on this.

[21:12] 发言人 1: Okay，again in this setup，if we are at a terminal state，then the value is no。Okay，if we are at non-terminal state，the value is one。Okay，so for any opponent states，we are looking at the worst value。

[21:30] 发言人 1: for any. Agent face or main agent，the one that try to win the game, I mean not your opponent yourself，is the one that I’m looking at。Any question about the agent master? Okay, so if now we can give you some examples, one of the one more example. Or simple enough for us to try is this big tactile, okay? So two player, one is playing a cross at one of the nine locations, the other opponent is trying to compete with you by putting maybe a circle at the other locations, so that if you join a line then you win.

[22:19] 发言人 1: Okay，so if you in today’s in this week’s tutorial to the notebook we actually implement this game for you to try this mean max search we will okay so

basically in this case whoever move works, you can treat that as the max agent would you who make a cross for this game. And for your opponent, we can assume that is the one was trying to make a circle, that trying to be this. So at first you have nine locations, after one make a move at the second move you have eight locations, and so on, so forth.

[22:56] 发言人 1: Right, so this is a perfect circle game. Zero sum. Okay, players will make turns alternately.

[23:04] 发言人 1: Okay，so now if we thinking about this game，it is actually a deterministic and the zero sum，right？So whatever you win，your opponent is going to lose。Okay，and it is deterministic，and two players actually take turns。

[23:22] 发言人 1: Okay，so now given this setup，we can develop develop a very very simple searching algorithm to find the. Uh, to find the best outcome and that is called the meaning max search algorithm. So what it does is that it actually computes knows meaning less value and the best achievable utility against a rational adversary.

[23:48] 发言人 1: So what it means is that when you’re trying to solve this game. You assume your opponent is rational means that he will compete with you all the time. He will do, he will take the action that will make your rewards, make your reward the words, okay? And this is basically what the medium max search algorithm will give you, okay? So if you’re thinking about this whole algorithm, you actually progress fairly easily.

[24:21] 发言人 1: So without showing you the suitable, we can actually take this graph to understand. Okay，let’s say the note developed by flu is actually my backstage。And the node you know by red is my meaning your opponent trying to compete it okay so now once you populate to the outcome to the leaf node of this tree if you are thinking from the.

[24:49] 发言人 1: Minimum the main agent perspective is trying to do the most damage to the back agent right so how about these two outcome the main agent is going to take the worst. out of the tooth. Right, so he will pick two instead of he will pick the action that will resolve into rather than pick the action that will resolve into eight.

[25:14] 发言人 1: Similarly for the other one, you will basically pick the action that will give you reward of five is six. Okay，so now from the perspective of the next agent at the root，he’s trying to maximize the whole thing，right？Maximize his own utility。

[25:32] 发言人 1: So now in the end，what he can get is five，because he only can choose between the action that will give you two or the action that give you five。Okay，so if you take a look，the best possible on hamburg，this whole game is actually eight。But the assumption is that he’s the product, the man’s ages assumption is that the owner is trying to compete with.

[26:00] 发言人 1: Is rational, so that he will do whatever the worst for the next day. The manager will do whatever the worst for the next day. Because that’s how the other opponent can win back.

[26:15] 发言人 1: So in the end, we have a lot. So if you put everything into a simple solution and algorithm, you can put them as this kind of recursive. Equivalence, okay?

[26:29] 发言人 1: So let’ s say we are the max agent, so we start with the max agent. So you will start with some value for this physical space, right? You will start with a space, and this space initially will have the value of inactive.

[26:48] 发言人 1: Because remember the state value, the value for the max state is something that you try to maximize. So the initial value should be as low as possible. So we assume that we.

[27:02] 发言人 1: Okay， and this states starting from this state， you were looking at what are all these child states that you can reach。And these child states will be controlled by my new agent. So when you go to this child state and fetch the value for all this child states.

[27:23] 发言人 1: You actually cause the mean agent who try to find out the mean value by every steps. Okay， so in this mean agent， we initialize this value to positive infinity because the goal for this mean agent is to minimize everything， so initially it has a worst possible value plus infinity， and this mean agent will again populate。Or his child state as well.

[27:52] 发言人 1: And he know all these child state again will be controlled by next agent because they take turns alternatively. So this. Value exactly by this function over here, so they recursively calling each other.

[28:07] 发言人 1: So the next agent look at when the next agent wants to calculate his own value, he needs to look at values of all his child states and all his child states belong to the other agent. So he needs to go to the other agent, the other agent will fetch. The values for this child notes, and for the other agent when he looking at his own child, that state will be again controlled by next medium.

[28:33] 发言人 1: Okay, so that’ s how they recursively together, and that’ s the whole thing becomes recursive, and this recursive ends. When you reach a leaf, when you reach a terminal. Okay， that’ s why you no longer needs to look at the agent anymore， you just pass passport。

[28:53] 发言人 1: Okay， so now to make things a little bit more clean， we can have a master node。Imagine this to be a controller, okay? Like a CPU master controller.

[29:05] 发言人 1: This master controller will define who starts first. For example, here you say the next agent starts okay, and then you will control the whole termination as well. Okay, so basically previously it is little bit ambiguous who starts first.

[29:26] 发言人 1: You can have a master agent over there says okay you start first, but then you move together. In the end you have some termination condition you can check. So whether that’ s not the game starts.

[29:38] 发言人 1: Okay， so that’ s how you can implement a mini-max search algorithm， and in this week’ s tutorial， we gave you one example in the book as well， on how this is implemented。Okay， so now if we thinking about a minimax actually example， it goes on as follows。Okay， so we follow the.

[30:00] 发言人 1: Now we know, it’s current with a particular agent, so in this case, it’s current with a maximum. Right, and we start with worst possible value because I need to see all these children to figure out actual value in the space. Without those, I initialize that to be accurately fit it because I want black by this.

[30:23] 发言人 1: So the worst possible is accurately fit it. Okay, then the algorithm says. I need to look at all my successive states, right? So for example, here I look at my successive states.

[30:39] 发言人 1: My success state is a mean state. Okay， and if this is the leaf node， I know its value， it is not in order for me to find out its. Mean value is to find out the value for this particular mistake.

[30:54] 发言人 1: I initialized this to be positive infinity, and again I look at all your child states that is basically. Let me share you this part. So, we start with the root.

[31:13] 发言人 1: We initialize the root to negative infinity. We now start looking at all its child states. Because the child state is not a leaf state, so I need to figure out if the value controlled by the mean agent, which is right here.

[31:29] 发言人 1: That’s why we start with this mean states, this mean agent node with possibility and start looking at its child node. If its child node is a leaf node, we have the leaf already available. Otherwise, we need to call this magic agent node.

[31:45] 发言人 1: Okay, so here it goes to this mean state, initially this with positive infinity. I do not, this is not illegal, so I need to look at its child. So after I really have a child, the child is a leaf.

[32:00] 发言人 1: Because of three, right? So the value of this channel, the value of this formula is three. Then I know for my.

[32:12] 发言人 1: In mean node over here, it has a value that has to be less than or equal to three. Because he taking a minimum amount. Right, so now the next one is12 doesn’t change.

[32:27] 发言人 1: The next one is8 okay? Only three available now. You know, without you for this day is three, you know for sure.

[32:37] 发言人 1: Okay， and you know for sure the value for this day has to be greater than equal to three because it’s taking a max of everything。Right, so I have already have to know the street, then whatever happened the rest has to be greater than me. So now, I move on to the second node.

[32:56] 发言人 1: This is a window, and I s- initialize this to be positive. Infinity, okay? This modulus, and also I need to go to its child now.

[33:06] 发言人 1: So the first is two, so I know this is less than equal to two. The second one is four, so nothing changed. The next one is six, I know this is two, this is three. Okay, so this two doesn’t change, this has to be greater than equal to three.

[33:23] 发言人 1: 首先还没有的。OK， so I popular again， this is a mean node， positive infinity， is not a leaf node， I go to his children， so this is14， that means has to be less than equal to14， ok? And the next is phi, so this change to phi equals phi.

[33:40] 发言人 1: As for me, so now it’s two. This is two and this is three. So the max out of these is three. Okay, so this is basically what the previous cycle is going to help you solve. Okay，any good questions? Okay，so if you are thinking about a minimax setup where you assume your opponent is rational，able to do the most damage to you by minimizing，so if you take a look at what will happen，if you assume that you play with the perfect。

[34:25] 发言人 1: Player，so if you play this game，you know the mean value is here is10，the mean value here is9，so the possible values，so even though there’s a very big potential outcome with a really really big reward of100，but given that your assumption is your opponent is a perfect player，then you will never be able to get this advantage。Okay，which is not the case if you are thinking about your component to be sometimes can make can make this day because because sometimes in reality. Your opponent may not be always perfect, he can make mistake, and that is our second part of this lecture expecting x.

[35:07] 发言人 1: So if your assumption changes from a perfect player who try to compete against you to a regular player who can sometimes makes mistake, then you can see from this. Okay, so now if you’re thinking about the efficiency of the, the efficiency here basically means what’s the space complexity, what’s the time complexity, does they guarantee to give you the optimum expenses. okay?

[35:35] 发言人 1: so if you’re thinking about a medium, it is nothing but aFS, right. that’s for a search. You go down, you trace down to the tree and you reach leave node, they will start properly in the value back.

[35:47] 发言人 1: It is just like a EFS, and more importantly, it is exhaustive. Because you need to visit everything that we know before you know the value for any known above belief, right? So you actually need to populate the whole tree. Okay, so uh pure belief back search algorithm is a always the worst idea as where you actually populate culture. Okay，so.

[38:28] 发言人 2: 然后。在。

[39:44] 发言人 2: 理论上是更需要的，不需要，不需要，不需要，这个这个这个这个，所以那个判断是判断的这个，就是。就是，就是你，假如说你现在觉得你走下来之后，这个地方，你发现完了之后，你要判断你是不是骗了他，和这个怕什么。这个建大决定的。

[40:33] 发言人 2: 这其实所谓的，我不知道为什么你说这个。就是每一个判断都基于这样的一个逻辑，对。就是我我我我如果是这样就是我们的判断是都是真的想。

[42:12] 发言人 1: うん。/sil.

[43:19] 发言人 1: So previously we have been talking about algorithmic programming and how we can basically. Not using this cooling technique to speed up the search of this very expensive meeting and search, but in reality that is not enough. Just the cooling itself is not enough.

[43:40] 发言人 1: So in most cases we are running a depth limited search. Just like the depth limited DFS, okay? We only run to store the depth limited, but as you may immediately trigger the fresh meal line is we need the value.

[44:00] 发言人 1: Of the leaf node to know the value of any non-terminal node, so without going to the end, how can you get this value? And this value is actually in many depths limited research, it is actually aspect. Okay, so depending on how good you can estimate, that is really determines the performance that we have。

[44:25] 发言人 1: So for example, in Eloago, people use a neuron number to estimate the value of any non-terminal states。Okay, and that turned out to be quite accurate. So in the end, the algorithm with data search algorithm works. Okay, so these are just some examples that tell you that as soon as you does not run to the end, all the error goes away. But if you want to run into the end.

[44:54] 发言人 2: you cannot solve it in your list data.

[44:57] 发言人 1: Okay, so if we try to look at these three. Different category of method, this is the original minimax which you just naively exhaustively populating the whole tree. This is an alpha-beta pruning where you populate partial tree, you prune some branch you cannot, you do not need to visit. Plus you not only pruning some part, you also run to finite depth and estimate value. Rather than going all the way to the root in every aspect of their life.

[45:30] 发言人 1: Okay, so for example, in terms of the optimal solution against the perfect components, both alpha beta purity and hitting axis guaranteed to give you the optimal, because you’re actually going down to the terminal states, right? And alpha beta purity only prove the branch that does not affect the solution。Okay, but that’s limited search, you has no guarantee, that completely depends on how good you estimate, 对吧?

[46:00] 发言人 1: Okay, and time complexity, you can see the original one is B raised to power M, and have a beta fruiting with a perfect loop bordering, you can save almost half, but for the desk limited actually that will need the mentality。Okay, depends on how many level you want to go back to。Okay, so in terms of space, black speed here, we are the same, okay, and.

[46:25] 发言人 1: For the algorithmic putting what you actually gave up is the partial solution, you never gave up the optimality. But for the depth limited search which allow you to solve on any realistic problem, you actually gave up the optimality. In general, we do not have a estimate of the value of any states.

[46:44] 发言人 1: That can guarantee you the optimal usually determine. OK, this is basically a holistic view for chess, basically a depth limited search is something we use to actually help us to solve. Being a chancellor. Okay, so this concludes everything I want to talk about for the minimax and alpha-beta pruning and just depth. This depth, this depth limited meaning that search any free question before we move on to the second bar where we now looking at not a perfect player but a player that can sometimes make these things. So any good question before we move on.

[47:36] 发言人 1: Okay, so now the basic our basic assumption changes a little bit previously that we talking about menu Max and8 of data basically we assume our components are just like a robot they never make mistake, okay, it basically every time as a cue in the max operations. I mean, you you. Take the worst operations that

do the maximum damage to the maximization, okay? But in reality, that may not be always true, okay?

[48:09] 发言人 1: For example, if you’re playing this human, even a human champion can sometimes make mistakes, okay? If that’s the case, then the only thing that changes here is if you take a look at this graph. Normally we have a brand note which are being agent right now we change this main agent to something called a chancellor.

[48:31] 发言人 1: Okay，so all these lower level kind of outcome can happen at some particular chance and some particular probability。Okay，so now this is the main idea where we assume the outcome can actually control or the outcome can happen as a probability。Okay, then we model this using the channel you see over here in green.

[48:58] 发言人 1: Okay, in red it means. You always pick the minimum in chance for green means every outcome can happen but with some probability with some chance okay so now we are going into another type of game which we call a stochastic game. Okay，when you have stochastic game，this transfer is really a convenient way so how we can model this game，like a monopoly，like anything that involves you story nice，you can be model using this stochastic game，this transfer。

[49:34] 发言人 1: Okay，so basically it is also very useful when you can，when you assume your opponent may not be always predict。Okay，sometimes he makes some random moves from time to time。That’s where you can use this child stop more。

[49:50] 发言人 1: Okay，so now if this is your assumption，if your assumption changes，for always take the mean value to all value happens at some probability。Excellent chance, then now what you want to find is not the worst case outcome anymore, it is actually the average case or the expectation or average was young, not always the worst case outcome. Okay，so now we are changing from mini max search to expect max，so the mean becomes expectation。That’s where the name Expecting Max happens, so this Expecting is basically the Expectation.

[50:37] 发言人 1: For the statistics side, okay? So basically what happens is the maximum still doing whatever is previously doing in the minimax, the chance node and like the pain node, but instead of always choosing, we choose the alpha based on its probability. Okay, so it basically also and we calculate its expected value.

[51:00] 发言人 1: So this expectation. You can simply understand that to be the weighted average, and this weight. It’s basically the probability.

[51:17] 发言人 1: Of each channel, okay? The probability for each channel to actually take place to occur, okay? So this is a basically a very general kind of introduction, but we will give you a more formal definition and more formal justification when we touch upon the walk of decision processes.

[51:42] 发言人 1: Okay，so we will show you this little bit of body，but now for now we just imagine instead of always pick the mid value，now we want to take the average weighted average of all the channel where the weights depends on the probability each node can actually have，so。That way is actually proportional to the

chance. Okay, so if you see here, everything remains the same, only thing changes this。

[52:14] 发言人 1: Compared to the minimax, the only thing changes is instead of simply taking the max or the mean value, I’m now calculating an expectation, and this expected value B becomes the value that we return. Okay, so just a simple example here now。These three showing right along this arrow is the probability for each note to actually happen, so the weight that average the weights is showing over here.

[52:50] 发言人 1: And now if you go with this equation, so what it says is that now I look over all its channel. When I’m visiting this channel, my V equals to my services V is zero, so my V equals to zero plus the probability happen. I’m basically following this.

[53:16] 发言人 1: Equation right here, OK? Times the probability times the value of the channel node in here, so B equal to four. This is for this node, then this loop again loop into the second node over here, so this becomes B equal to four.

[53:35] 发言人 1: This is B already changed to four, right? So four plus. ONE THIRD TIMES TWENTY FOUR, so equal to twelve. This is when you visit the second node.

[53:46] 发言人 1: When you visit the third node, it becomes B equal to twelve plus one six times negative twelve, so negative two. So B equals ten. Okay, so if you put together, it’s basically a big song。

[54:02] 发言人 1: Okay, the way is coming from the problem there。Okay, then for now just thinking about if someone just will provide the probability. Okay, when we’re learning about rock of decision process, reinforcement learning, and we at this patient network, we will show you how you can get these probability.

[54:22] 发言人 1: But for now, just imagine the probability will be given. Okay, so in case where we do not give you any probability, for example in this case we are not showing any probability to you, you assume that every node have equal probability。So for example, in this case we have one third, one third and one third.

[54:45] 发言人 1: Okay, it’s just a convention。Okay, so now something came to your mind after we learning the mini max algorithm。The first all of work we learn is how we be making more efficient。I truly sometimes。

[54:59] 发言人 1: Right, so now imagine can you burn it anymore? Okay, so if we calculate the expected values of this part, which equal to1 plus4 plus3 equal to8, right? This node is8, so we know this node is greater than or equal to8。

[55:19] 发言人 1: Okay, so now we want calculate for this。Okay, let’s assume that we know there are three nodes here. We just don’t know whether we need to go to them or not.

[55:30] 发言人 1: So I know this is one third times two plus one third plus some value here, B one. Plus one third comes out of E two here. Will this validated and on your name? Can you hear me? Any idea?

[56:00] 发言人 1: Can you pull or can you predict?

[56:09] 发言人 1: The general answer is no. Because this value, this B1 within two can take any arbitrary value, and there’s no guarantee that it has to be greater than equal to a S. But if you can assume.

[56:27] 发言人 1: That every node takes some value within some bound, for example, if you say that every node has a value you estimate, every node to be have a value you can see on the chart, then you can roughly calculate. Okay，so in general，the hazards now，because every now is. 理论上 Speaking，你可以给任何价值。

[56:51] 发言人 1: Right, but if you have some prior knowledge on what the value of every node can actually take place, for example here we assume that it can only be zero to ten, then you can calculate this value. Okay? So again, you can also run that’s limited expecting max.

[57:10] 发言人 1: Okay, so the only thing changes right now is previously you need to run all the way to the depths to the terminal state. To find out its value and the back propagating back, right? But now you’re only going to hear and say you have one two three four.

[57:30] 发言人 1: You say that’s limited. You go to four, you can only run four depth. The way you do this is by estimation.

[57:42] 发言人 1: Okay，this estimation may not be correct，but it has to be at least informative for your algorithm to actually work。Like we’re police，as I previously said，nowadays many of people using neural network to estimate this value。Okay, so all these trick that we previously seen for being max can be applied, okay? But not straightforward.

[58:09] 发言人 1: Like in the polling case, we need to have some assumption. On the value of challenge before we can prove and expecting next also you need to somehow have an estimation. All the value of that email before you can actually round to this. Okay，so any question on this.

[58:29] 发言人 2: part?

[58:33] 发言人 1: Okay，so now I want to very give you a very quick recap on little bit of these probability which is needed later on for the mark of decision processes as well as for you to understand the things。Okay，these are very very. All the next one and that battery stuff, okay?

[58:54] 发言人 1: So, if you’re thinking about the probabilities, what I want you to think about is. Random variable that represents an event which outcome is unknown, like flipping a coin. Okay, flipping a coin itself, you can model that as a random variable where whether a hell, whether a tail or a head is the outcome.

[59:16] 发言人 1: Okay, make sure you do that. Okay then. A probability distribution is basically an assignment of debates to its outcome. Okay，so what he does，imagine this distribution，this probability distribution through a function，through a mapping，that if you model this as a random variable，then you know if what’s the actual weights or what’s the actual probability for this random variable to equal to any particular outcome。For example, this probability distribution can tell you if you flip the coin.

[59:55] 发言人 1: Too many times, many many times you know the probability equal to the outcome of a head is roughly50 percent. Okay, send the whole thing together becomes your probability distribution. Okay, so let me give you an example to make things a little bit clear.

[60:14] 发言人 1: Let’s say now. The event that you are trying to look at is traffic on the highway. Okay, and you didn’t know this.

[60:23] 发言人 1: As the variable T, which says whether there’s a traffic, whether there’s a traffic jam. Okay, so it’s outcome of three cases: where it has no traffic, where it mild traffic, or heavy traffic. Okay, and what we mean by this probability distribution is basically the following three things: this is a discrete.

[60:44] 发言人 1: Probability distribution，it tells you there’s twenty five percent chance where the highway has no traffic，there’s fifty percent chance that there will be loud traffic，and twenty five percent chance that you have very heavy traffic。Okay，these are just some alignment of determinologies。Okay，when I’m talking about random variable probability，this basically what?

[61:09] 发言人 1: So when we talking about the expecting max，that’s basically why。Okay，so the chance node is your random variable here，and the outcome is basically this child node，and the probability distribution gave you the probability along every edge。Okay，so this is basically what we mean。

[61:31] 发言人 1: So some of the facts or laws that any probability has to follow, first of all, every outcome within a probability or within a formal probability distribution cannot be that. So the every outcome should have a positive or zero probability to happen. You cannot have a negative chance of happening.

[61:55] 发言人 1: Okay，and if you sum up with all the chances，it has to be someone。Okay，and if we are getting more and more evidence，the probability itself can change，and this is the main part of one of decision processes later on，okay，and which is called a condition probability。Okay，so normally.

[62:25] 发言人 1: We know the traffic, the probability of having traffic on the highway is for example twenty five percent. ok 嗯。If now we have more evidence. Or more observation, for example here the observation the evidence we have is the current time is morning rush hour8:00. Then the probability of a heavy traffic on the highway can increase to forty-six or sixty percent.

[63:00] 发言人 1: So what you really about this probability is what’s the probability of having a heavy traffic on the highway given that right now the time is8 PM? Okay, and we call this a conditional probability。Where after this vertical bar is our evidence，our observation。

[63:24] 发言人 1: Okay，so again we will talk about the reasoning and updating this probability later. Let me talking about part of this new process and motivation. Okay, so now with all these，you can imagine from the probability science the expected value.

[63:43] 发言人 1: Or this is the expected mass, the expected value we are calculating is basically a function of the random variable in its average sense weighted by. The probability distribution over itself. ok 首。

[64:00] 发言人 1: Now, thinking about this traffic, this probability of traffic of traffic of hiding cases. Now our variable is better, you have non traffic, mild traffic or heavy traffic. Imagine this to be the expecting max cases, right? You have three child node.

[64:19] 发言人 1: This is your chance of life. And this is equivalent to the character random variable we are talking about. You meant it is to be known, is to be mild, is to be calm.

[64:34] 发言人 1: Okay，and let me tell you，this has forty percent of the time going to have fifty percent。And put it okay, so the expectation with this case is basically you can make the average if you use this. Some of them are basically going to this.

[64:54] 发言人 1: Okay，and here is basically you the time，right? The time here。is basically the value in here, so20,30, and60, which tells you if there’s no traffic, you will take20 minutes to the airport, if there’s a lot of traffic, take30 minutes, heavy traffic, take60 minutes.

[65:15] 发言人 1: Okay，so what does calculate this on average? How long it take you? From your place to the airport, and this average is average over.

[65:28] 发言人 1: All the possibility of the traffic, all the highway, waited by the actual probability of each traffic type will come. Okay，and this is what we mean by expectation，and this is basically what the chance from this compute。Okay, so.

[65:53] 发言人 1: In reality, when you’re looking at, when you’re solving a moral this problem as a probability, uh, program. Let’s say reasoning, we need half this problem right? So some of the very simple model that we can use is beautiful model.

[66:09] 发言人 1: You say that everything happened in equal size. That’s basically what we mean by the uniform distribution. Okay，just by rolling a dice，every site, flip a coin，every site，equal probability。

[66:21] 发言人 1: Okay，and this model can be actually more sophisticated if you want to actually represent the real world. Okay，so now like I said，for this，as far as this cost，as this cost is concerned，we assume that we will give you this formula。Okay, so whenever you solving exercise or when you are encountering a problem from midjourney，for example，we will provide you this。

[66:49] 发言人 1: Okay，在 later part of the course，we will show you how you can estimate and how you can update this from there。There you go, okay? So here is just a simple question that if now you know your opponent is actually running at that too.

[67:08] 发言人 1: Using the results eighty times eighty percent of the times and moving randomly on this. So one kind of search, three search algorithm. So it means that you are playing with the opponent.

[67:21] 发言人 1: Who is being rational? Running a depth to meeting Max is basically telling you that he will do whatever does best for him. Okay，at80 percent of the time，and for the remaining20 percent of time，you will have randomness。

[67:37] 发言人 1: Okay，so in this case，I think the answer from you should be obvious。You should run this chance note，where80% of the time you will pick the big value，and20% of the time you will pick everyone in that。Okay，so this are just some.

[67:56] 发言人 1: Some simple real-world cases that to show you that in reality. This is something that you can be used to formulate this problem, so here we directly tell you what’s about right? So you have to do a shot over there.

[68:12] 发言人 1: Okay, so next a very important. Concept I want to get across is this model. Okay, so if you’re thinking about the part one, the first hour versus this part we have been talking about, the method is roughly the same.

[68:32] 发言人 1: The only thing changes is your wrong assumption about your opponent. In the first hour when we talking about minimax expect that the minimax and elevator booby we are assuming your component your opponent is perfect player who never make this thing. And in the second part of the lecture, we assume that your opponent is not perfect.

[69:00] 发言人 1: We can make this. Okay，so if your assumption matches the real world situation，you will have the best outcome。If your expectation does not match the real world situation，you may feel very miserable。

[69:18] 发言人 1: Okay，so let me give you the following example to actually show you what can happen。Okay，so now we can imagine。You are running a packman case, running a packman game, learning controlling this packman to eat the ball, at the same time not getting killed by this ghost.

[69:37] 发言人 1: Okay，and now when you’re designing a search algorithm，a gameplay algorithm to control this packet。You can have two different mindset. The first mindset is you assume the ghost is perfect.

[69:57] 发言人 1: the ghost is not going to act randomly the ghost will kill the pack by all of them. Okay，that’s all by meaning by meaning by expert，so you are being pessimistic。You assume that your goals is going to have.

[70:15] 发言人 1: The second Pacman is you assume your Pacman, your ghost, the ghost opponent you are competing against, can make a mistake at some time, and sometimes they just make random move. You may not want to kill the Pacman all the time. Similarly，this is。You be angry, so this is your assumption.

[70:42] 发言人 1: And this is what the reality, okay? So if your assumption that the ghost is going to be perfect and the reality is your ghost is perfect, if you are running a minimax algorithm, you can win all the time. Okay，you can make all the time, and you can have a good average score。If your expectation you assume the ghost is adversarial is perfect but in the end your ghost is actually not adversarial it’s just random you will still be. All the time, and maybe even have a slightly better score because your course is not acting rationally all the time.

[71:30] 发言人 1: Okay，now the next case is where it becomes dangerous is if you assume the ghost is going to behave randomly from time to time，but in reality your ghost is always the same，your ghost is really perfect every time you try to beat this

you you will lose in both of the cases and have your really bad score。Okay，and the last case is where you assume your ghost is going to run. Behave from time to time, and between reality yes your process also behaving randomly from time to time you will be next one so if you see the diagonal is where your assumption matches the reality thus in most cases you win and have pretty good score especially in this part.

[72:22] 发言人 1: Okay，in all diagonal is where your expectation does not match the reality。If that happened for some cases，it can be dangerous，especially when you are trying to be overly optimistic。You think they are doing bad stuff, but in reality they are doing perfect stuff. Okay.

[72:44] 发言人 2: that’s why I need now.

[72:47] 发言人 1: Okay, so one of the example is this backgammon. So I don’t want to show you much more details, but instead for running a minimax, we have run an expected max to assume that. Things can happen, and some chance, and in the end we can solve this game pretty nicely.

[73:06] 发言人 1: And the way we solve this is using temporal differences algorithm. So when we talk about reinforcement learning, I will show you this in detail. But here is just an example that telling you going for minimax to expect the mask and allow us to solve some of the very complicated problem.

[73:25] 发言人 1: 但是。Okay，so the last part I want to cover for today’s material is the concept of utility. Okay，so we have been seeing this utility happening again, again and again，particularly the whole essence of this course is about maximizing expected utility.

[73:46] 发言人 1: Okay，so so far in this four。Like first we have been telling you all kinds of method how you can maximize. Right，for a simple tree search problem, you can run。

[74:00] 发言人 1: DFS DFS，uh，uniform cost，that the uh that’s limited DFS, excess right? If you have a no cost of course，then we tell you how you can use uniform greedy search at this one。Okay，well，if you do not care about the path, you just care about the final outcome，we show you the CSN，the constraint site version time。

[74:24] 发言人 1: Right, how you can use boot uh network target server to solve it. Now we show you if now instead of having one agent, you have two who try to compete against you, how you can maximize your own score by meeting max of the beta moving and expected max. These are the different ways how you can maximize.

[74:45] 发言人 1: We have been showing you so far. Next to lecture, we will show you two other method how you can maximize. Now this expectation, which is very quickly show you the expectation is basically the weighted average over all this probability.

[75:00] 发言人 1: Distribution right, and we look elaborate more when he goes through this work of decision processes and reinforcement. Okay, so the final part, the utility part, I want to talk about little bit more. Okay, so first of all.

[75:21] 发言人 1: In this case, we are talking about the expected. You are average what’s your opinion? So you need your question even wonders why not meeting next which talks about the worst case.

[75:34] 发言人 1: Right because. 因为。月牙湖。Of co- of course, the worst case gave you the best guarantee in terms of not to fail.

[75:48] 发言人 1: But in reality, this may be a little bit too conservative, and the germ, it will jeopardize the actual outcome you get. So, it is actually in reality, we actually. actually. A better assumption or assumption that better suited the global situation is the weighted average of costs probability rather than the worst case.

[76:14] 发言人 1: Okay，so that is the choice that we made to say that a rational agent should choose the action that maximizes the expected utility，not all，not the worst case utility all the times。Okay，so some of the follow-up questions that you may have is where do we get these utilities，and do we actually have such utilities exist? Right, for example, now if you’re thinking about just some recent example that our man have tried to solve this, let’s say now we want to train a robot to work like a human being.

[76:51] 发言人 1: Then how do you define a utility to measure how good or how bad it is like you? For example, if you’re thinking about an art design, you’re designing an art piece, and how do we define it properly? Okay，so all these are valid questions to do，but something I want you to be a little bit here，especially valid or expected utilities performance。

[77:15] 发言人 1: Okay，so you can imagine utility to be some kind of functions。Okay，so for example。Let’s say this is your.

[77:28] 发言人 1: Original search，and this is your original value。Okay，and let’s say now instead of using the original value，I can model the utility to be square of. Okay, so that means instead of I have zero or plus zero, instead of forty, I can have sixty thousand.

[77:49] 发言人 1: So every of these value here has been squared to itself, just like you put in the distance, right? You can square it. Okay, so these. Transformation of your utility works when you are looking at worst case, so if you are running a meeting next.

[78:11] 发言人 1: To the left and the two right, if you’re doing this you have the zero here, you’re doing this you have the twenty, so this has strong. So your optimal solution is to go this path. And go this far, this is your solution.

[78:29] 发言人 1: If you’re running a minimums to the right, you have a zero here and a four hundred here and a four hundred here. Of course, the value itself changed, but the actual action you took is still the same. Right，so if you have a monotonic transformation。

[78:52] 发言人 1: Means that before transformation it has sort of water, after transformation it has remain the same water, the order doesn’t change. Then the worst case minimizes reasoning maintains that doesn’t matter what kind of you simply function you have, as long as it is monotonic, it is order preserving, that it

does not change the outcome of your minimax. But if you’re thinking about this to an expected utilities now, you’re not doing any.

[79:24] 发言人 1: Mean value anymore, this in all these becomes chance. Okay, so here you have twenty。So in here we have twenty five assume equal probability okay so in here we have a hundred.

[79:41] 发言人 1: So here you have how many650? Right, so the max out of these two you have forty-five miles of these two. THIS IS THREE HUNDRED, SO WE HAVE THREE HUNDRED.

[79:56] 发言人 1: SO THE ORIGINAL SOLUTION IS GOING THIS PATH. NOW THE SOLUTION IS GOING THIS PATH. Right, so now if you are now thinking about expecting acts where the actual value matters, even monotonic transformation of the utility may not actually capture the logical solution.

[80:19] 发言人 1: So what this tells you is how to imagine the utility. Will change the outcome because remembering this course, most of our method is trying to maximize the expected. Okay, so what I want to show you here is you need to be a little bit careful about the utility you choose, okay, because that in general defines the outcome that you get。

[80:47] 发言人 1: Okay, so the utility are the functions in summary utilities, the functions from the outcome。You can imagine this to be a particular final state of the world, and this utility。It is the function that maps these outcomes to some real values, which can describe what you want or what the agents want.

[81:10] 发言人 1: Okay, so if you want this action board, you should require. Okay, so. Going back to the previous question, where is this utility function?

[81:21] 发言人 1: So for a single game playing, it could be win or lose, like plus one or negative one. It’s very easy to define. And actually there’s a theorem by von Neumann, he shows that this is any rational preferences can actually be summarized by utility function.

[81:42] 发言人 1: And maps this outcome states to a particular value. Okay, so in reality, what we do is we as human being, we control this human being. Okay, so we specify what is good and what is bad, and we do not let the agent。

[82:04] 发言人 1: Okay, change annually。Okay, and basically what we want is we set up this utility such that we hope。What we want, the desired behavior can emerge after maximizing this expectation.

[82:22] 发言人 1: Okay, we do not directly tell what you can or cannot do. So let me give you a concrete example: when you’re using, for example, AI algorithm or reinforcement learning algorithm. To try our autopilot driving vehicle.

[82:38] 发言人 1: We do not tell the vehicle that you cannot do this, you cannot do that. We formulate everything into a reward such that when this reward is maximized, we will do the desired behavior. For example, if you’re hitting a tree, you get the reward minus100.

[82:56] 发言人 1: If you’re hitting a human, you get a minus a million or even. In this key, at the bottom of the globe we have both factors. All these being factored into a reward function such that automatically maximizing this reward itself can encourage some good behavior.

[83:15] 发言人 1: The things that you need to be careful is we never let the agent touch this reward usually. So if you are looking at some of these news on the blog, you might have heard about this reward hacking or thinking about this recourse of self-improving, that’s also very relevant here. So wherever you want there’s a good behavior to emerge, you need to make sure that you have a very well-developed, you’ve put the function that your agent do not easily trick.

[83:47] 发言人 1: 就那鼻子一样。Okay，so all these are something I want to cover for today in terms of the material is concerned，so let’s take pretend。And resume around eleven thirty, then I will use thirty minutes or so to show you the project as well as some of the tutorial material this week. Okay, so we will do back around eleven thirty.

[85:39] 发言人 2: 是啊不学习就机智了。

[86:09] 发言人 2: 老师说这两个是分开的。

[90:19] 发言人 1: 你是从哪里看到的?

[94:23] 发言人 1: Just like last week, this week’s he told me has two parts. One is a two little. For you to actually see how many nice works on this detail games, this has been implemented for you so you can run it and see how it works and check that against the ideas from the lecture as well.

[94:48] 发言人 1: This is we expect you to run this and understand how it works okay at your own time. There’s also a local copy of the duplicate of the download is the same. Second part is a reading exercise.

[95:04] 发言人 1: I suggest two questions. One is on the meaning of x, alpha beta moving. The other part is on the expected max, so give a try and try to solve it yourself to see if there’s any confusion part and anything.

[95:22] 发言人 1: So before next week’s lecture, we will post the solutions similarly to the previous case. We also will post the solutions for you. 嗯确认。うん。

[95:39] 发言人 1: I think. YES. So next week we will post the solutions for.

[95:51] 发言人 1: But this part, okay? So what is this? You just try to solve this yourself first, okay? Then we will show the solution. Any question on this part?

[96:02] 发言人 1: Okay，if now let’s move on to the passport again，this will be30% of your final grades，and the most differentiated30% I will say，and I give you a quick walk down last week，you will mainly have three parts。One is the proposal. Okay，so the most small will take about twenty percent of your total project rate，so in total that will be about six。

[96:32] 发言人 1: Percent of your total grades for this course, and you basically need to submit one page. PDF summarizing. Okay，who is the team? May what you want to solve? And why do you think you can solve it?

[96:52] 发言人 1: Basically，you need to do a little analysis to find out what on the benchmark，on the data，to see is anything that’s. Help you actually carry out the project. So when we looking at this one page, we need to know who is your teammate, what is your problem, and we need to get some sense that you understand what you are doing.

[97:16] 发言人 1: Okay, so you have going to the literature, going to you have search online to see these are available resources, so we need to see that, okay, and then you understand where you’re from. Okay，that is due on the fifteenth，so I think October。Okay，so after you come back from the National Day break，we will have another lecture，and at the end of the following week，is this when this is due?

[97:43] 发言人 1: So about three weeks，you can form a team，so we encourage you to form a team with three teammates。Okay，you can have up to three together，over number up to three。And I think that’s a more manageable.

[98:00] 发言人 2: Nice.

[98:02] 发言人 1: and better distribution of work。So one or two is fine，but you have a little bit more work based on our experience，and we do not individually create everyone。So the whole group will receive one grade, and that is to negotiate non-negotiable.

[98:23] 发言人 1: Okay, so we do not want to see team members coming to us saying that he does all the things or some metrics, not anything at all. That’s not our problem. YOU WANT HIM AND YOU MARRY.

[98:40] 发言人 1: Okay，so one grade for every group，then double appreciate you。And then one or two months out, we will ask you to actually one three weeks or one month prior to the final due day of the project, we ask you to submit this. Monster okay, this monster have no.

[99:07] 发言人 1: Control the format or anything. We just want you to show us that you have started. For example, you can give us one page showing that you have done this literature review, you have come up with all these method and summarize what you have found so far.

[99:21] 发言人 1: It’s fine. Or you just show us a code that says you have implemented this, even though it doesn’t run. It’s fine.

[99:30] 发言人 1: Okay, we just want to see you have started. That’s all. And we do not create this base of merit, we create this as a possible film.

[99:38] 发言人 1: If you submit, you actually show you have started and gave you four months. So it’s a binary. As long as you get started, you will get graduates.

[99:51] 发言人 1: Okay，and the last part is the problem。Which you need to submit a report, not long only four to six page, I think four page maybe and code. Okay，so we don’t want you to elaborate many things，just focus on what you have done。

[100:15] 发言人 1: Okay，what’s the new knowledge or what’s the new thing you have，does it? In the code, okay? You can upload this if you can upload to Google Colab, you can send a link.

[100:26] 发言人 1: Otherwise, you can upload to GitHub or GitT or wherever works. In many that’s fine, but our expectations as follows. We will download health, follow you with me.

[100:42] 发言人 1: Exactly，and if it doesn’t work。微波那个地方。Okay，if it doesn’t work，you do not have to grade on the coding part。We will exactly follow your reading.

[101:00] 发言人 1: Your requirements says you have to install this, install this, install this. Even you tell us you only work on this Linux version, we will install a Linux version exactly as you said. But after we follow that, it doesn’t work and it’s not our problem.

[101:17] 发言人 1: We would not be one. Okay，so if your code doesn’t compile，you do not get a grade for the coding. Code is not the full mark.

[101:29] 发言人 1: It says we distribute the points to each one of the part. For the coding part，you do not have that part. We follow exactly on how you tell us to run, based on your reading.

[101:41] 发言人 1: Or if you send a Google Collab, it’s easier because I just run on a Google Collab. If it doesn’t work, we will not be up. Okay，then some of the formatting instructions，okay，we hope you to follow so that it’s easier for us to break，okay，and. Okay，first of all，any question on the logistic.

[102:05] 发言人 2: part?

[102:07] 发言人 1: 你打游戏干嘛？Any problem on the schedule, on the expectation, on the grading, everything related to the project. So if you don’t know where to find.

[102:26] 发言人 1: If you go to the here project structure, click, it will take you to this page. Otherwise, you can directly go to page. And you can go to first project, you can see all these. ok. Any great question before I move on to the topic?

[102:54] 发言人 2: What product can we choose?

[102:56] 发言人 1: Yeah, that’s the on the private topics. I mean, any question on the on the first one? I hope you like. 嗯。Okay，if not on the project topic，so what kind of project are you.

[103:12] 发言人 1: Anything related to AI spot, which I find it very hard if you can find a project that do not relate to AI. Okay, so pretty much anything is fine. If you have no idea, we gave you three, but right now there’s only two.

[103:28] 发言人 1: But we hopefully by the weekend we can finish the third. Yeah, basically. Right now, there’s two show you over here.

[103:37] 发言人 1: The first one you have already seen from lectures when we talk about heuristic. We show you that how DeepMind Google DeepMind used AI to write AI algorithms to write heuristic functions. Okay, and that is the first topic.

[103:53] 发言人 1: Let me show you here. So if you read this here, we briefly. Introduce what the problem is, exactly what we mean by how we can use large model AI to automatically design algorithms, design AI algorithms, and we provide you some code to guess right.

[104:14] 发言人 1: So Google DeepMind released their paper, but they did not release the full code. So we we provide this for you. Let me see, right? Here is the Google’s big mouse official representation.

[104:33] 发言人 1: And this is our reinforcement, which contains everything for you to run. Okay, and we gave you some directions on where your problem can happen. For example, you can apply the same method to different type of problems, like to the routine scheduling, etc.

[104:56] 发言人 1: Also, you can enhance it. how can we make it more efficient. And.

[105:06] 发言人 1: There’s another application of how can you be simply making this to search algorithms, can you make it to machine learning algorithms? And many arts, I think this is basically the first trend using AI to automatically design AI. Okay, here’s some programs.

[105:27] 发言人 1: You need to have access to large language model API, or you need to have a local large language model running yourself. For example, if you’re running a MacBook, I know how to run it on my MacBook. Otherwise, you can... you need to have like GLM, KIMI, or DeepSeek, for example, DPI software.

[105:49] 发言人 1: So normally we can provide API to you. When we offer the course in Hong Kong, but here there are some issues which we cannot offer. Okay, so for you to carry out this, you have to figure out your own API。

[106:10] 发言人 1: Okay, there’s some university policy and legal issues here, we cannot give that idea。Okay, so if you want this, you will have to, for example, the easier way to get this guy according to what, chemo from where, it’s not much, you know。Um, it’s one hundred, one hundred, ninety, okay, get this.

[106:34] 发言人 1: So I’m very sorry, but due to the university policy and regulations, we cannot provide that API. Any question on that? Was on there? Use the AI to design an algorithm.

[106:50] 发言人 1: ok. Then the second topic is concerned about how can you model a real or complicated problem into a search problem. You are able to model something as a searchable problem, and we are giving you a target right here, which is called a secure invert.

[107:11] 发言人 1: Okay, so right now you’re using this large language model like chatgpt, deepseek or chamber library for your book, everything you type。Hydrodynamically can be read or memorized by these large amount of problems. Right?

[107:28] 发言人 1: So there’s a demanding topic on how can you increase. Is data. At the same time, do not change the model and hope it can work.

[107:39] 发言人 1: And that is called the humble, that’s the whole before we. Uh, The FHP encryption for large life one。So right now, I’m telling you, we are giving a project showing you how you can formulate this problem as a search problem。

[107:59] 发言人 1: So. To begin, we show you one example on how can you do this for a convolutional neural network. And we provide you with the code and everything for you to try.

[108:14] 发言人 1: So we provide the problem formulation and the code on this combination. Then what we want you to try is try this to transformers, to large language models. Okay, so basically you need to fall, you need to do something very similar from CNN to Transformer, make it to work on the transformer。

[108:39] 发言人 1: Okay, so if you go to the files. On this project, on the same topic, on the security risk, we provide you with examples and example populations for you to guess. And for this, you do not need a large.

[109:00] 发言人 1: You completely work with your laptop or any computer. Okay, any question on this talk? So how do you formulate a secure inference as a search problem and action solve? It's the second problem.

[109:23] 发言人 1: And the last project idea which we have not showing right here, we will provide another project along the heuristic. So let me teach you the heuristic. Heuristic means how can you estimate how good or how bad this is.

[109:40] 发言人 1: Right, so now if we are not thinking about the state, can you estimate how good or how bad a large micro model is? So right now every large family model takes. months even to years and millions of dollars track.

[109:59] 发言人 1: Can we develop a heuristic function without training the large language model can predict how good the large language model is? Okay, so that is also very challenging, and we will give you something to start. This field is recently started. We have some work along this lines, so we will provide you with future more introduction and call for you to get started.

[110:25] 发言人 1: How can you predict a new number before even you get started? Okay, so those are the three preescribe the topic that we have in our mind, because we have work on this, and we have some code, and we know the research landscape a little bit better better, so if you have question we can support. So these are advantage of predefined topic.

[110:51] 发言人 1: but you are more than welcome to choose your own topic. anything is fine. If you want to work on reinforcement learning, it is fine.

[111:02] 发言人 1: If you want to work on computer vision, it is fine. If you want to work with video, etc. Okay, any question on the topic?

[111:17] 发言人 1: Okay, so I said that this year we cannot provide any GPU or LLM API due to policy issues。Okay, so。In case in the end, so these are some questions we receive from previous years as well.

[111:36] 发言人 1: So in the end, if you develop something that's good and we see a potential of developing this into a formal journal or a conference paper, then we will work with you together to make it right. For example, we will have graduate students help you. We will actually provide you with GPUs and provide you with resources.

[111:57] 发言人 1: Okay，we will have regular meetings，so that after the course project，you can actually take this group course project into actual submissions。Okay，we have this every year in the past two years，and from the course we have submissions to these top conferences，and this year we have one one accepted papers from this。Okay，so just keep going，you can try。

[112:21] 发言人 1: Okay，so that’s all of these。Partnership logistics，things to remember is it takes thirty percent of your grade。Okay，you need a group maximum three members，every member receives the same grade。THE FIRST ASSIGNMENT RELATED TO PROJECT IS DUE IN THREE WEEKS, WHERE YOU NEED TO SHOW ONE PAGE PROPOSAL, TELLING ME WHO IS YOUR TEAMMATE, WHAT’S THE TITLE, BASICALLY WHAT YOU WANT TO DO, AND SHOW ME THAT YOU UNDERSTAND WHAT YOU WILL DO.

[112:58] 发言人 1: Like we have do some research, you know what’s the benchmark, what are your competitors, etc. in this one. increase. Okay，any question?

[113:18] 发言人 1: Okay, if not, that’s all I prepared for you for today. Okay, so I will be here for questions you may have, otherwise you are able to go. Okay.

[115:03] 发言人 2: 对呀，我刚看到了。

[115:09] 发言人 2: 哦。哎呀，玫瑰花。

[115:20] 发言人 2: 对，有很多这样的例子，就是这个小问题，就是有些时候就是没有这个面，就是这个信息，所以他们推测你要求解的问题。假设我们现在理解的问题是一个大模型的价格，未来涨价。想放到设备上那我们不能真的在去练预测我们会有具体呃就。中心小圆和顶部的象征。

[117:09] 发言人 2: 好，就是 PPT 的问题。嗯，呃，我们这个 PPT 还要再改吗？还是等我们这个项目做出一定成果了再再，还要再改啊？还要再改，就是还没给老师看，还是？嗯，等等他把那个商业部分填上去之后，给老师，给那个，发给那个校长看一眼。ok.

[117:40] 发言人 2: 能通那是最好的。商业部分点上去之后再给我，我还要那个换电动化之类的，还要提醒一下，不然的话只是一张张放过去就很难看。嗯。

[117:57] 发言人 2: 嗯那就我回头做啊只要负责 P E T 部分的对你们还要搞那两个智能体就是如果还有其他的就叫我们搞一下嗯。

[118:14] 发言人 2: 然后跟你呃跟组员说吧，就基本上每周每周每个人进行一次汇报嘛，嗯就是对自己的成果汇报一下啊，如果没空的话，就我们就线上会议。嗯。啊，好。那我先去吃饭了。

[122:29] 发言人 1: 这。

[124:13] 发言人 1: 真的。

[124:20] 发言人 1: 自己。